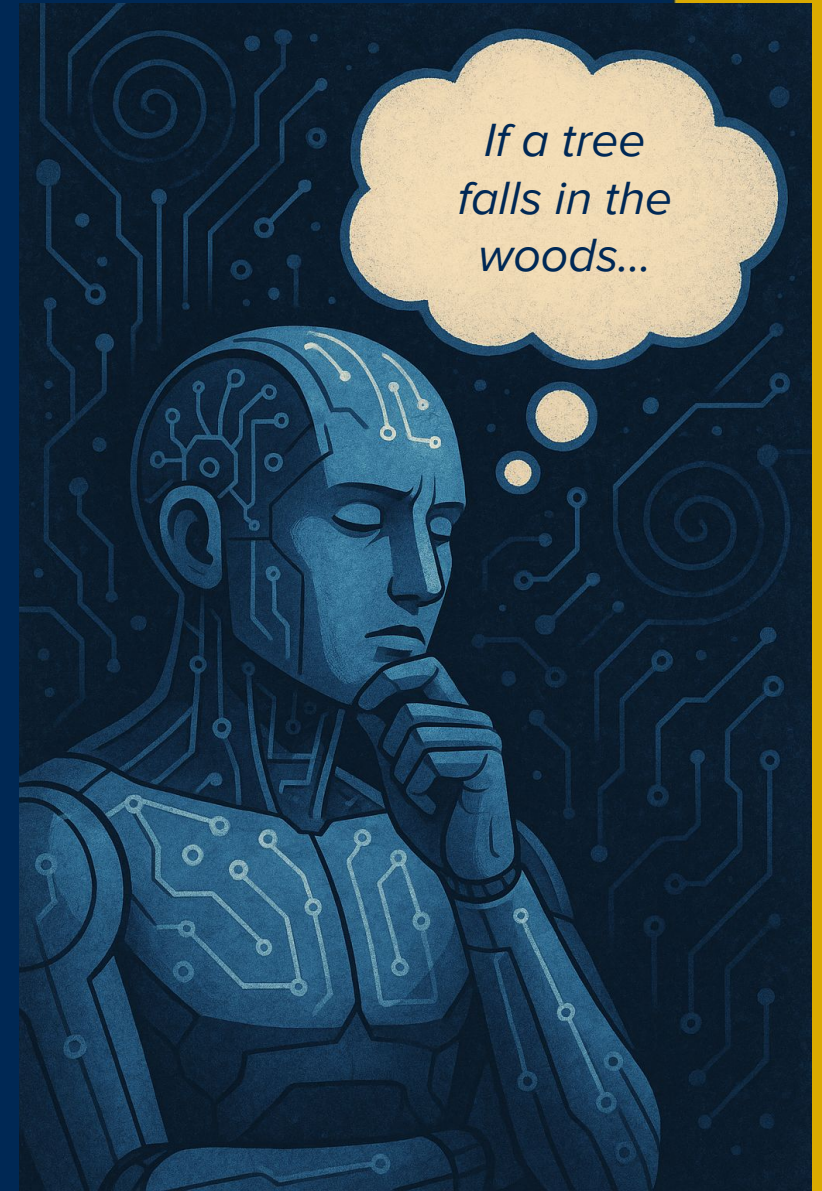


UCDAVIS
DataLab

Data Science and Informatics

How AI Thinks

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14 November 2025



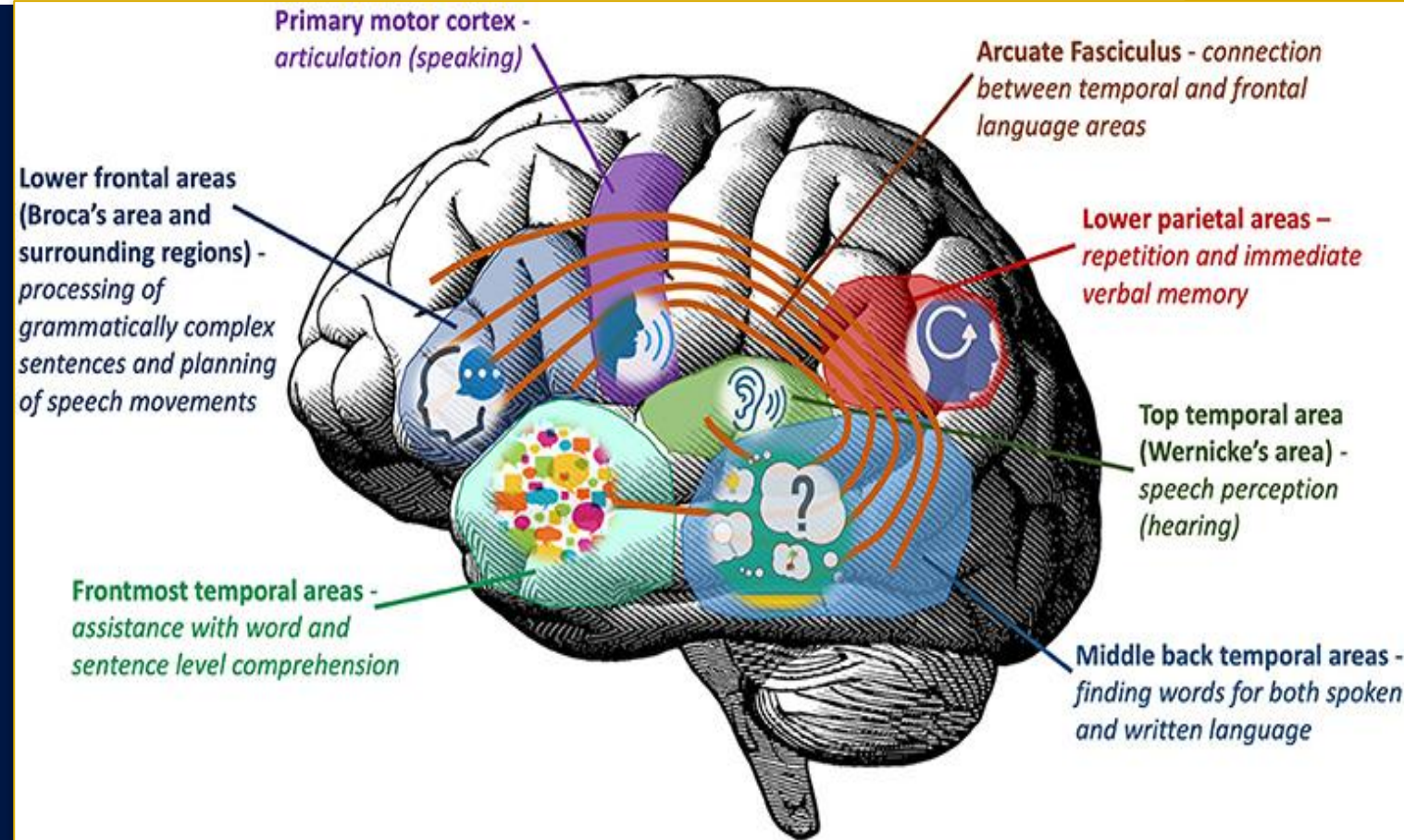
Some Preliminaries...

What we won't cover:

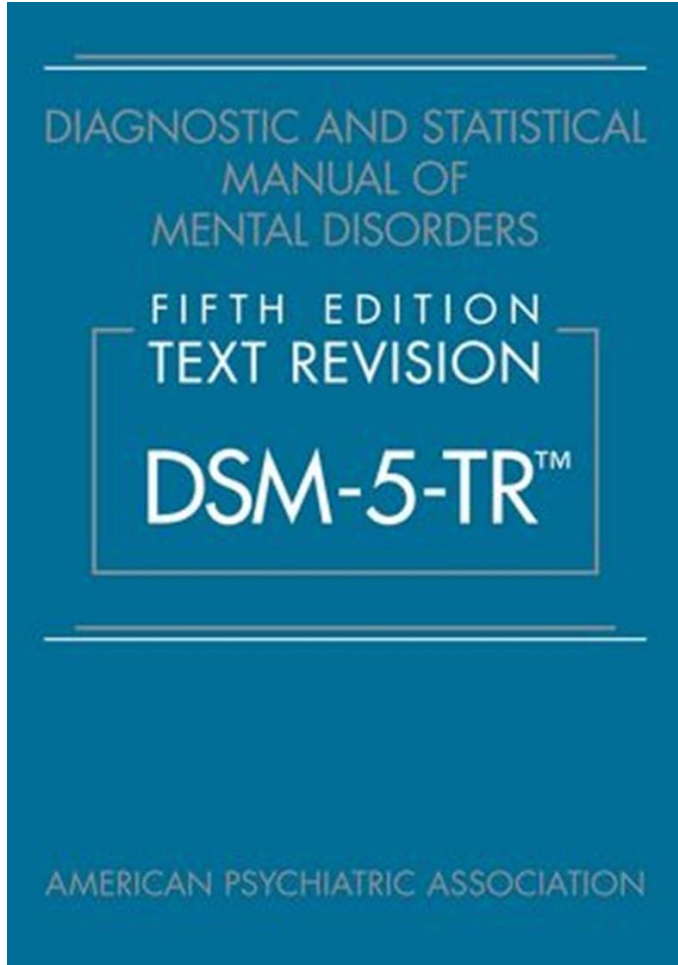
- Algorithms or math
- Coding or implementation
- Specifics of various implementations (for example RAG, etc.)

What we will cover:

- Theories of human cognition and language that AI mimics
- Brief history of evolution of AI
- Conceptual understanding of the components of AI systems and how they operate to mimic human thought



Domains of Cognitive Function



Attention

- Ability to direct focus to multiple things at once and choose where to focus attention and what to ignore

Executive Function

- Control and coordinate other cognitive abilities

Learning and Memory

- Ability to record information that can be retrieved when needed

Language

- Including semantics, grammar, and syntax

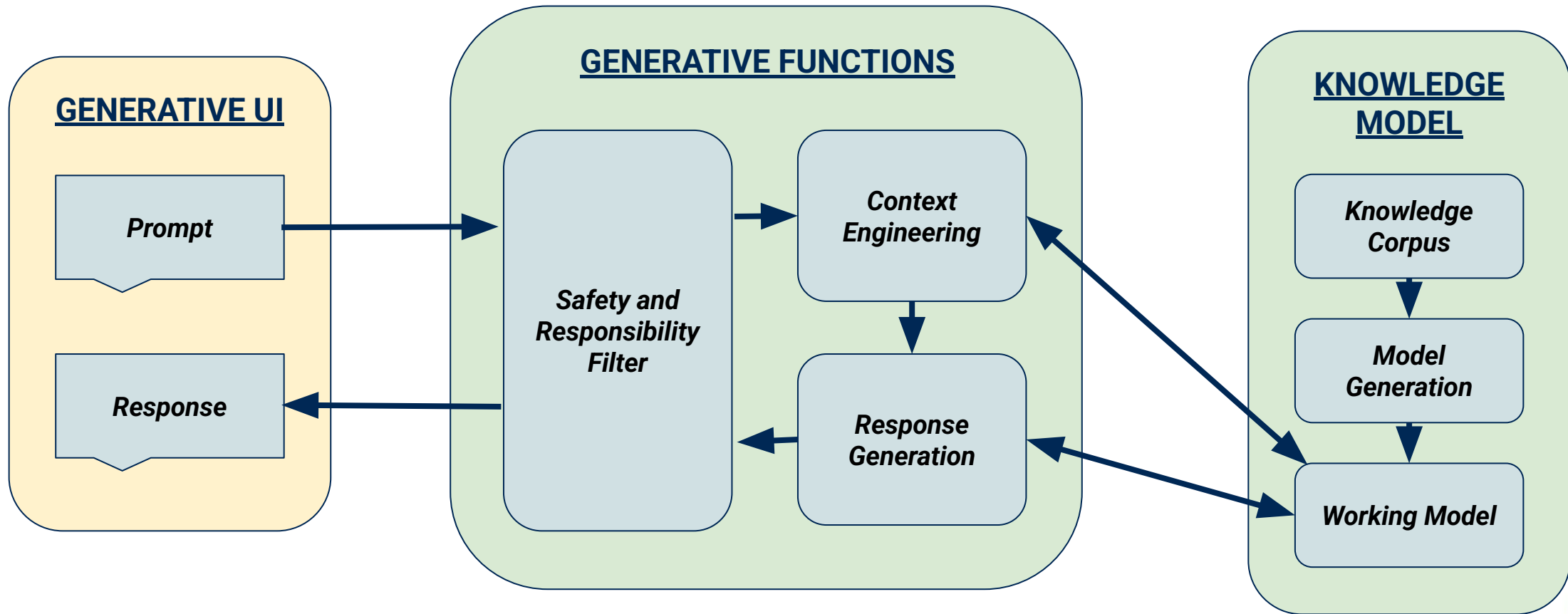
Perceptual Motor Control

- Ability to coordinate movement in response to physical surroundings

Social Cognition

- Ability to process, remember, and use information in social contexts

Cognitive Domains and Basic AI Workflow



- Attention
- Executive Function
- [Language]
- Social Cognition

- Learning and Memory
- [Language]

Learning and Memory



- North Torrey Pines Road and Genesee Ave, UCSD

Holographic vs Localized Information...

Localized Memory

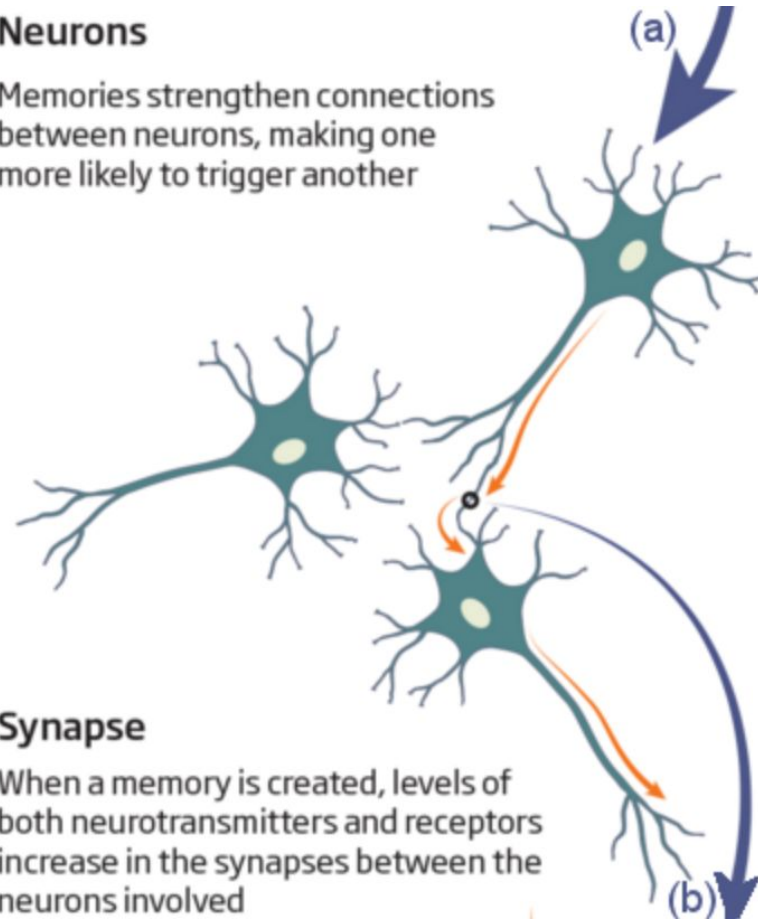
- Defined collections of neurons and synapses responsible for storage and retrieval of discrete tokens of information.

Holographic Memory:

- Definitions of discrete token are distributed across the entire vocabulary of tokens
- Each idea is defined in relation to all other other ideas

Neurons

Memories strengthen connections between neurons, making one more likely to trigger another



Synapse

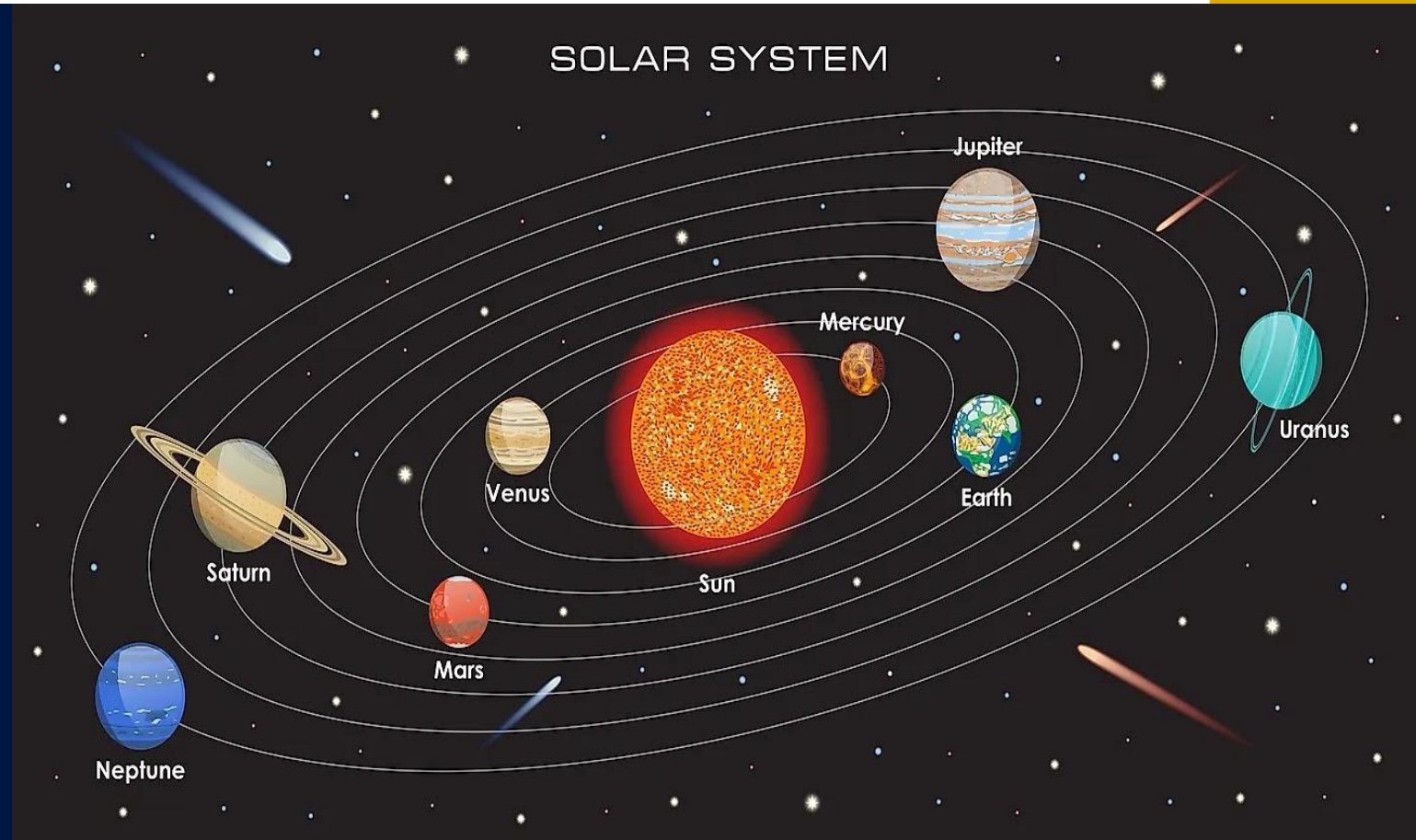
When a memory is created, levels of both neurotransmitters and receptors increase in the synapses between the neurons involved



How it Works

The Vector Model

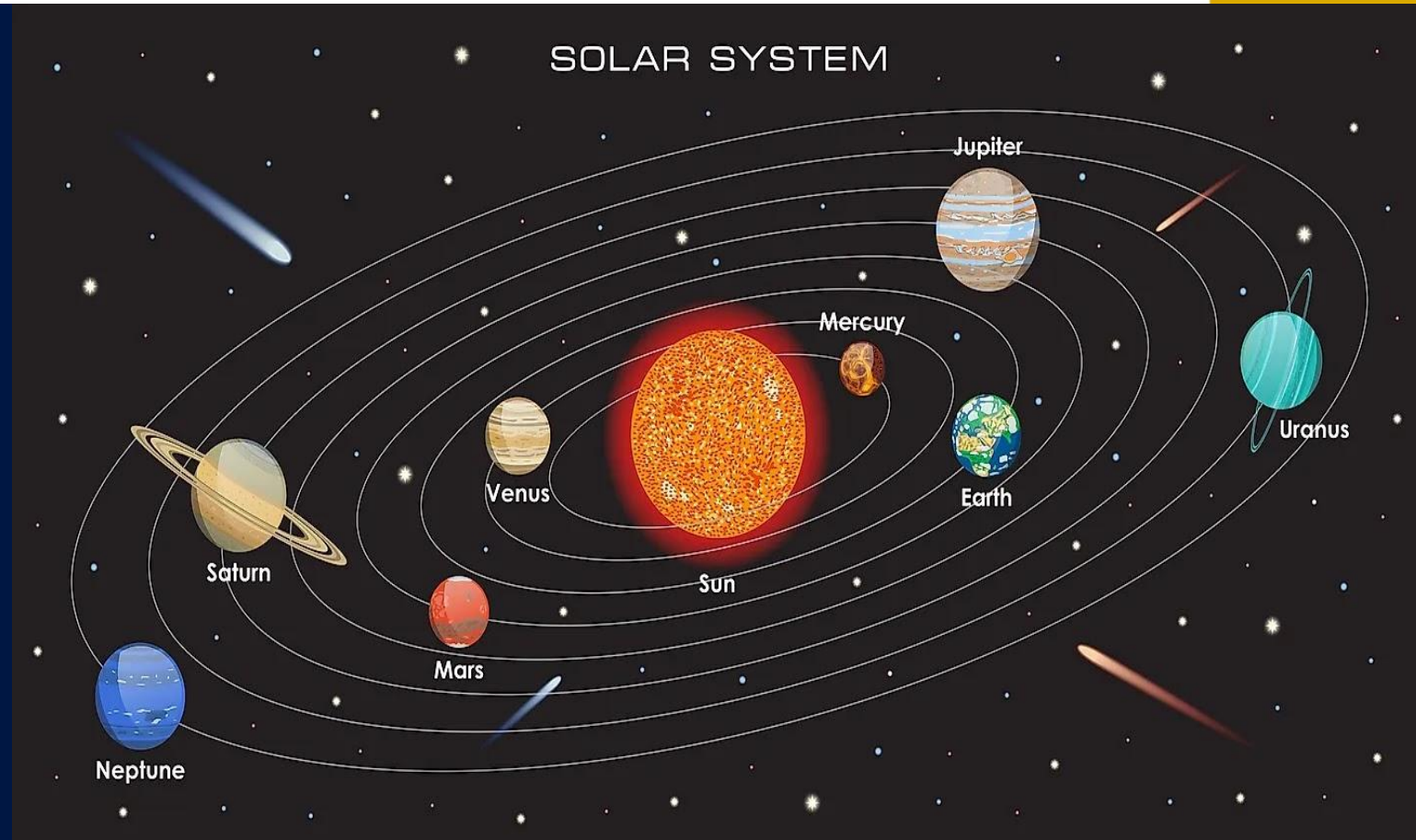
- Neptune = [2, 3, 1, 6, ...]
 - Mars = [2, 1, 3, 1, ...]
 - Earth = [1, 1, 0, 4, ...]
 - Jupiter = [1, 1, 0, 4, ...]
-
- Each number [element] in the list [vector] represents a mathematical relationship between that planet and all other planets.
 - For Example, element 1 might be the average distance between all planets, element 2 the median, etc.



How it Works

The Vector Model

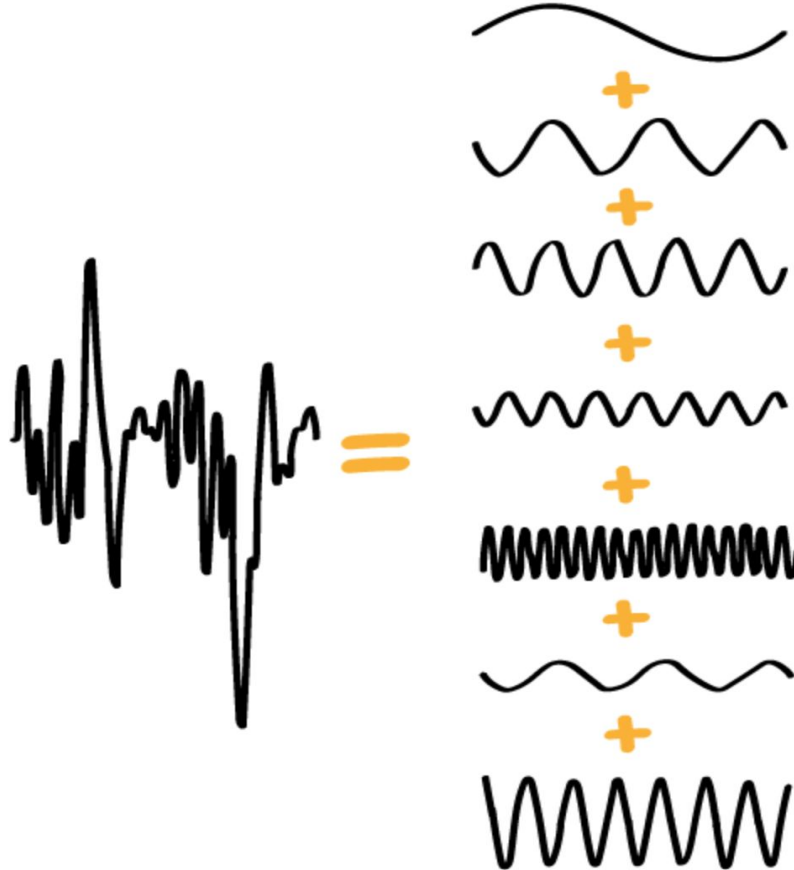
- Neptune = [2, 3, 1, 6, ...]
 - Mars = [2, 1, 3, 1, ...]
 - Earth = [1, 1, 0, 4, ...]
 - Jupiter = [1, 1, 0, 4, ...]
-
- Each object's vector is called a "Dense Vector".
 - The collection (matrix) of vectors is called a Tensor



How it Works

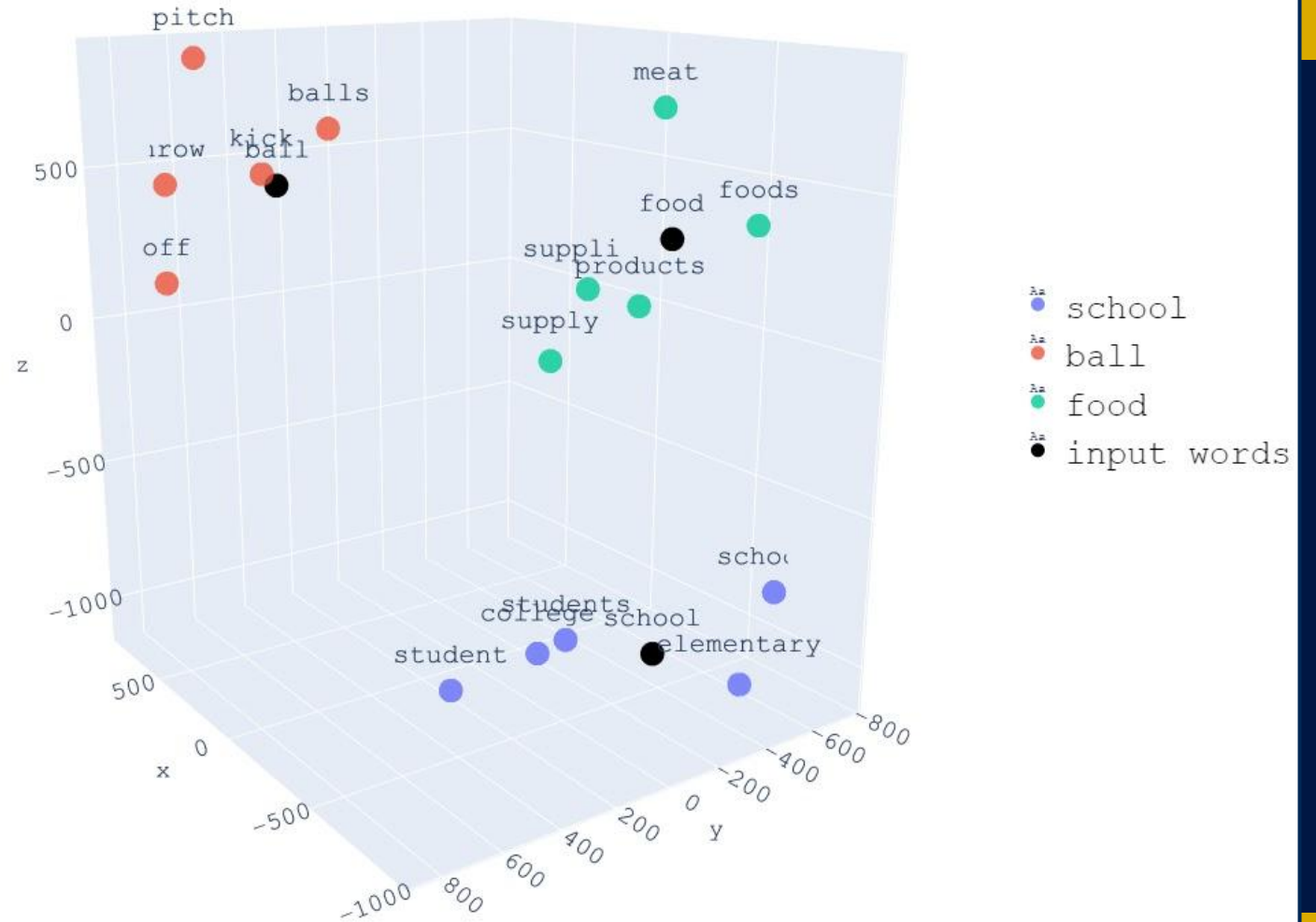
There is no want of knowledge respecting what is wisest and best in morals, government, and political economy, or at least, what is wiser and better than what men now practise and endure. But we let *I dare not* wait upon *I would*, like the poor cat in the adage. We want the creative faculty to imagine that which we know; we want the generous impulse to act that which we imagine; we want the poetry of life; our calculations have outrun conception; we have eaten more than we can digest. The cultivation of those sciences which have enlarged the limits of the empire of man over the external world, has, for want of the poetical faculty, proportionally circumscribed those of the internal world; and man, having enslaved the elements, remains himself a slave.

Percy Bysshe Shelley



How it Works

- **Tokens that appear closer to each other in the training set, have more similar vectors**
- **Tokens that have more similar vectors, are closer to each other in meaning**



How it works



“You shall know a word by the company it keeps.”

J.R. Firth, 1957

How it works



“You shall know a word by the company it keeps.”

J.R. Firth, 1957

Carl Likes Cats

Cats Carl Likes

Likes Carl Cats

How it works



“You shall know a word by the company it keeps.”

J.R. Firth, 1957

***Carl Likes Dogs When they
Stay Over There***

How it works



“You shall know a word by the company it keeps.”

J.R. Firth, 1957

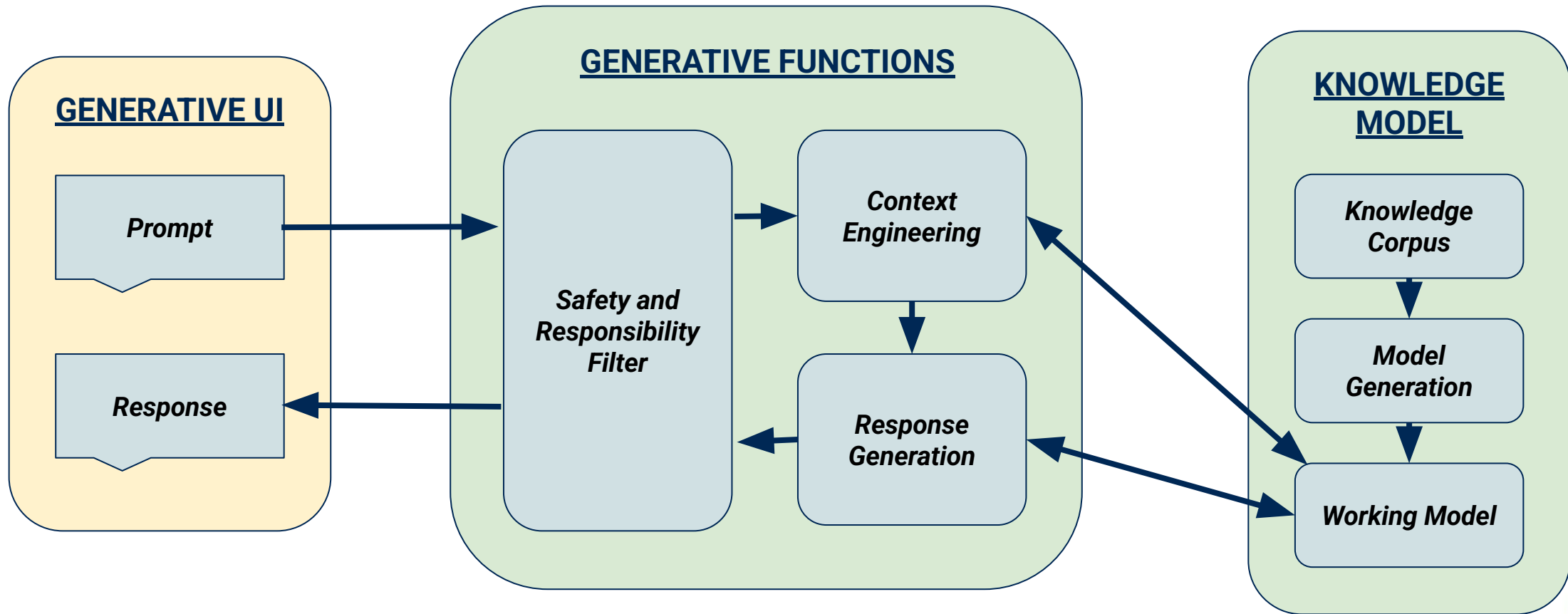
Carl Hates



Carl Likes



Cognitive Domains and Basic AI Workflow



- Attention
- Executive Function
- [Language]
- Social Cognition

- Learning and Memory
- [Language]

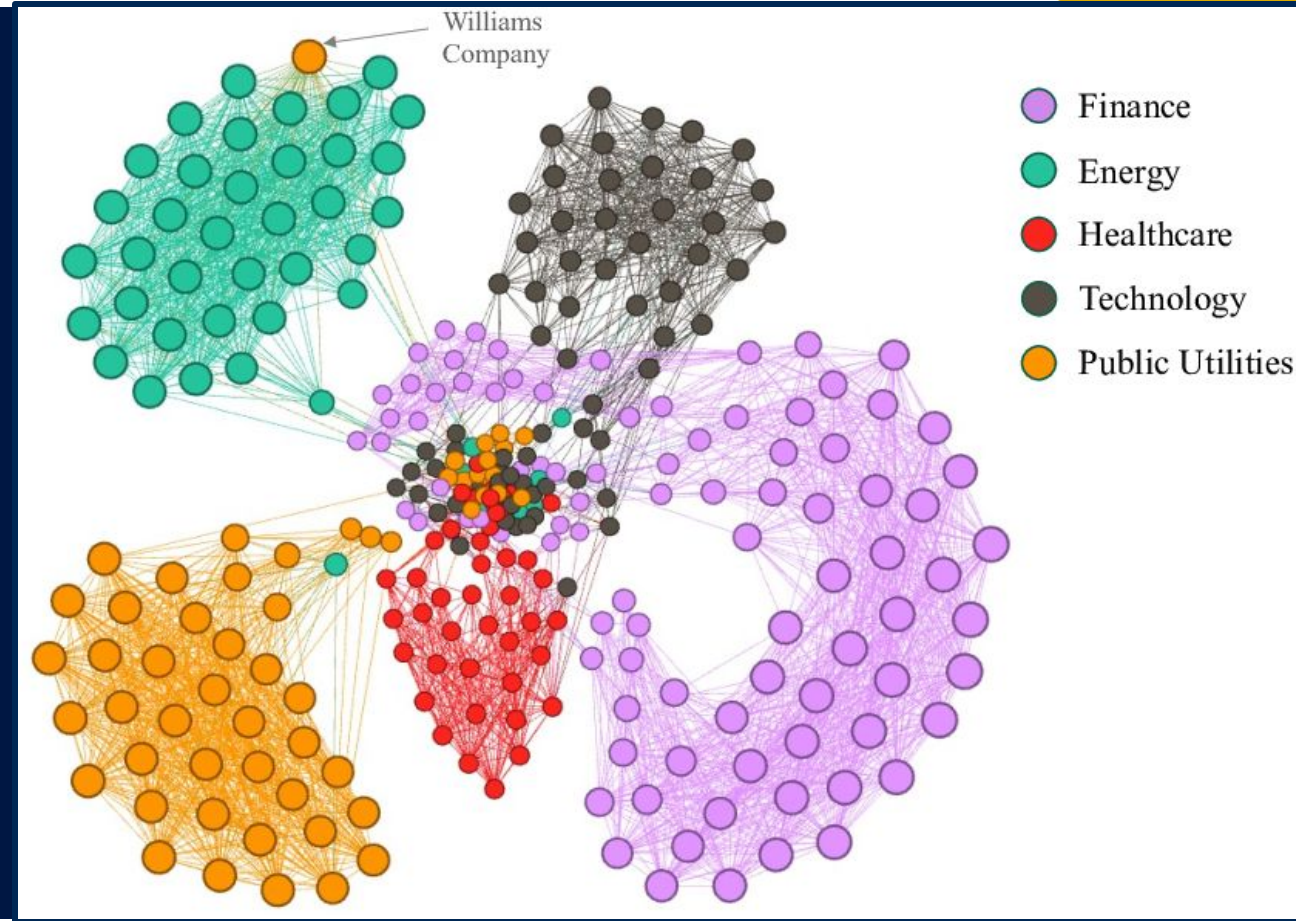
A Quick Detour: The Black Box



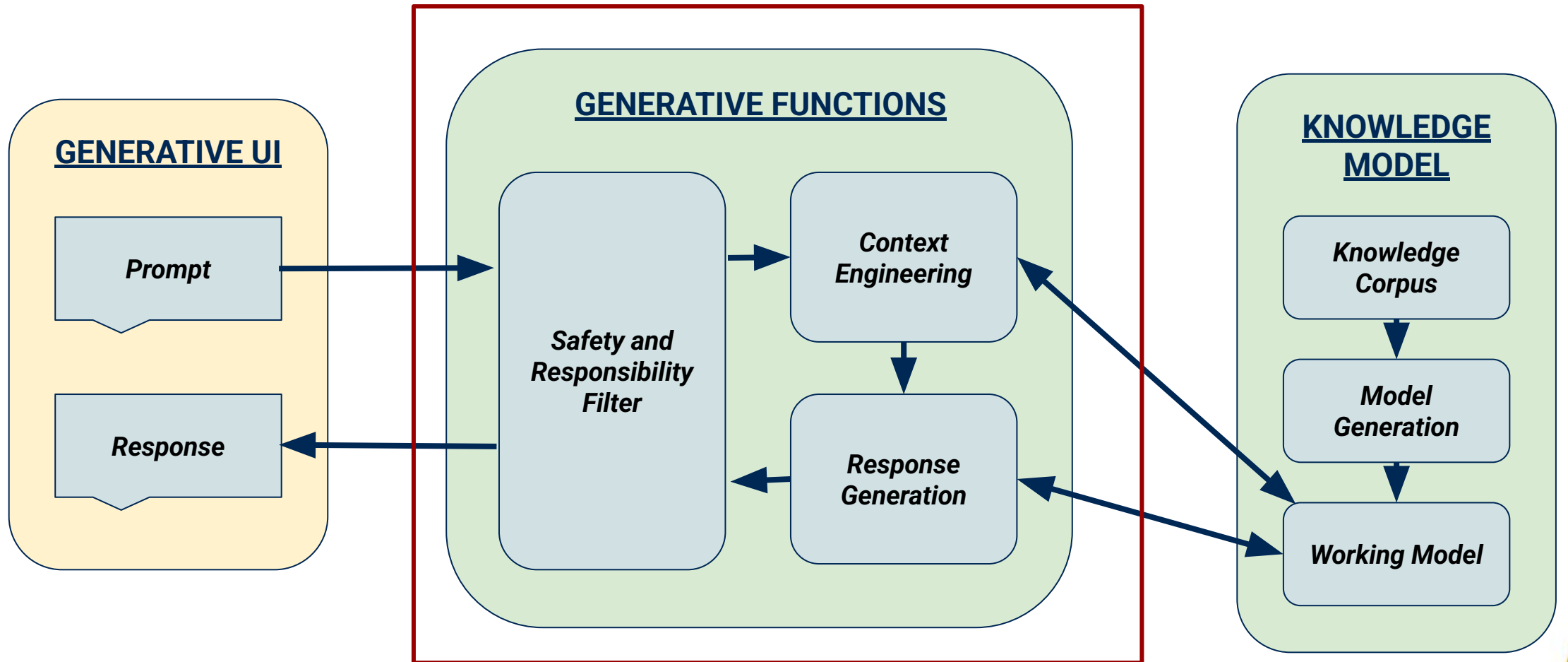
Another Detour: Model Based Research

Many research and search and retrieval tasks involve direct engagement with the model:

- Clustering
- Classification
- Search based on document similarity
- Corpus analytics



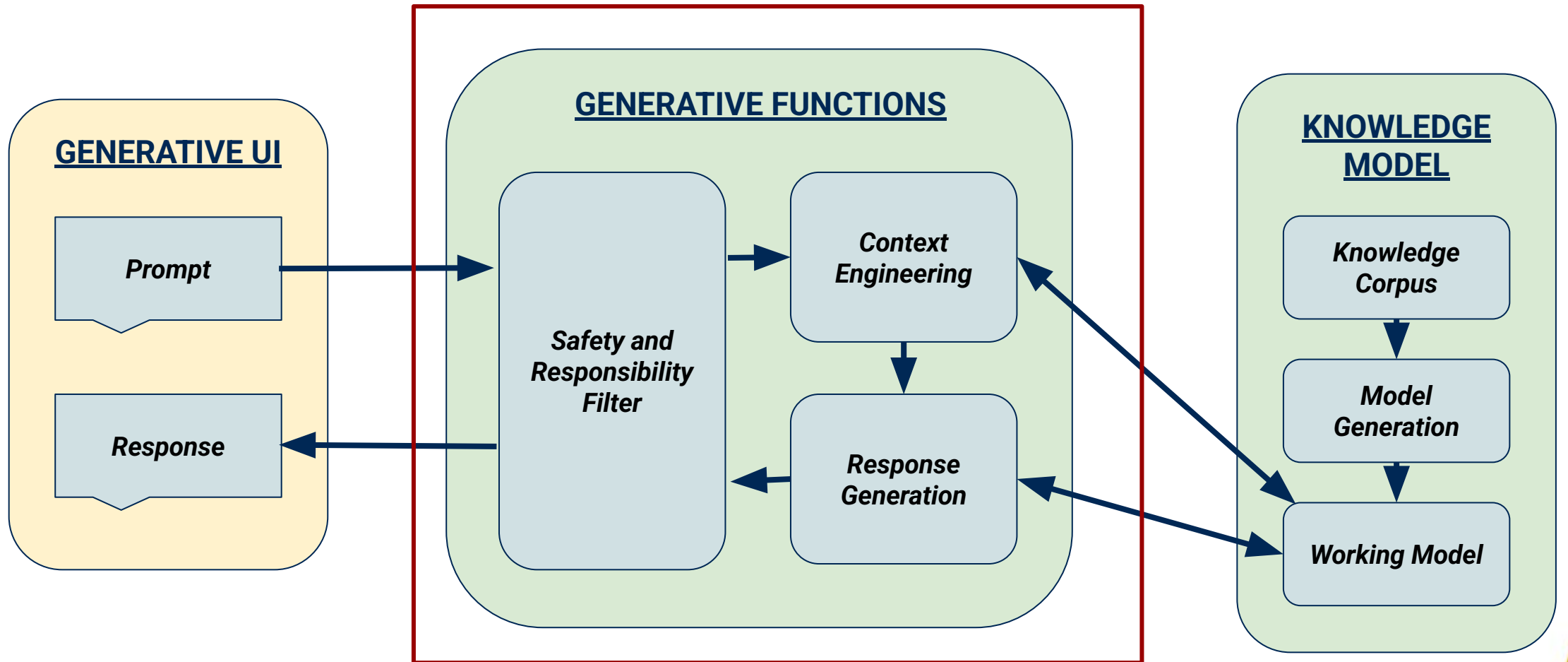
Let's Talk about Inference



- Attention
- Executive Function
- [Language]
- Social Cognition

- Learning and Memory
- [Language]

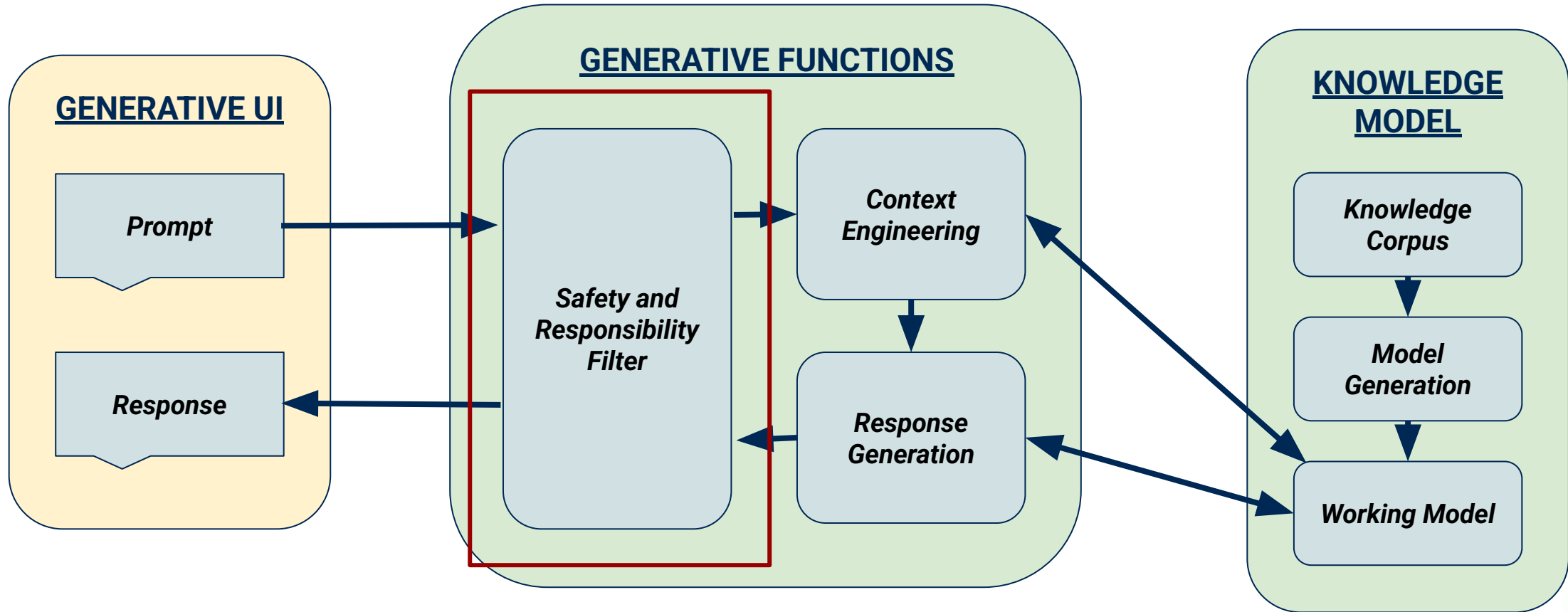
Executive Function



- Attention
- **Executive Function**
- [Language]
- Social Cognition

- Learning and Memory
- [Language]

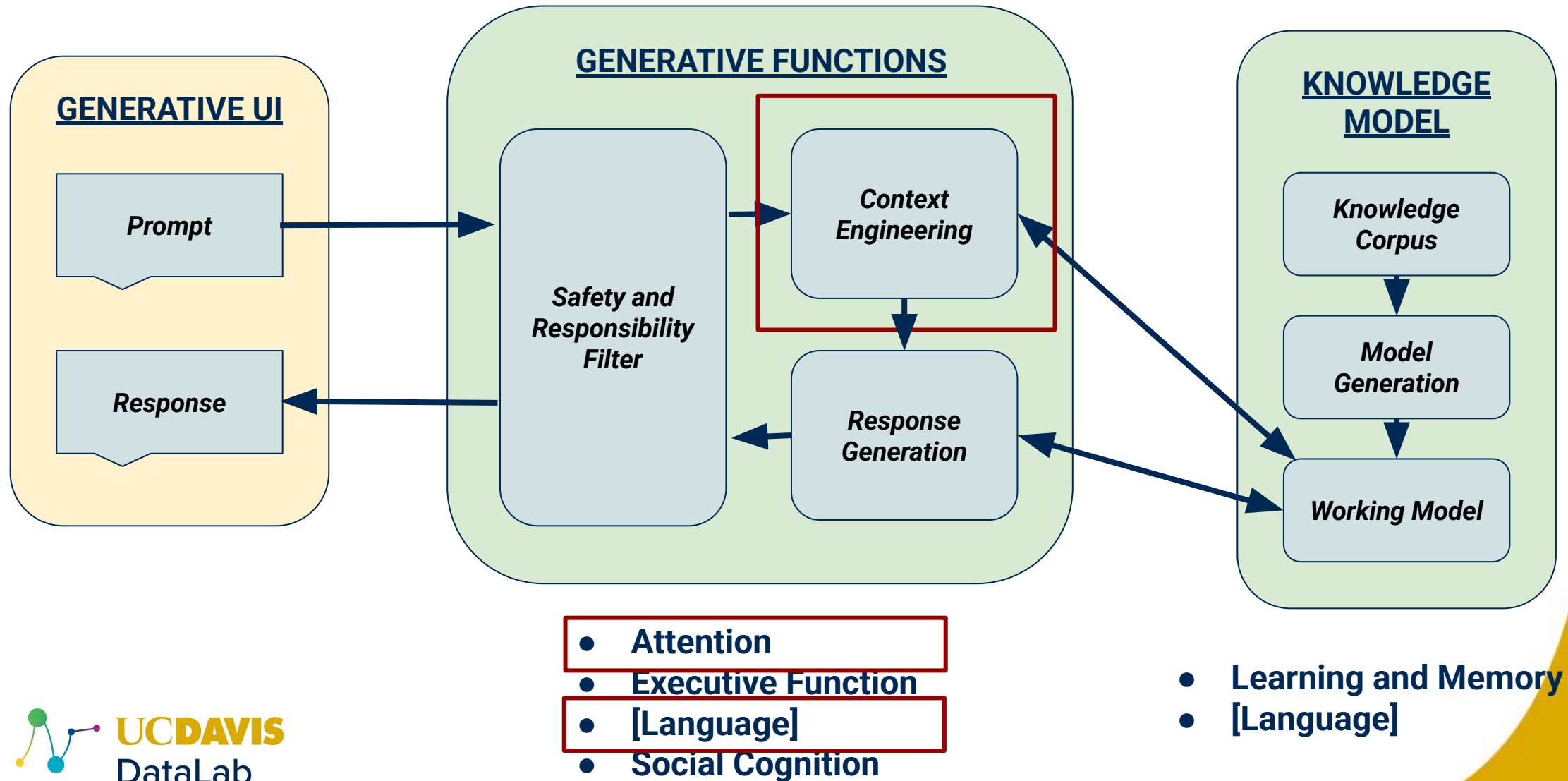
Safety and Responsibility



- Attention
- Executive Function
- [Language]
- Social Cognition

- Learning and Memory
- [Language]

Context Engineering

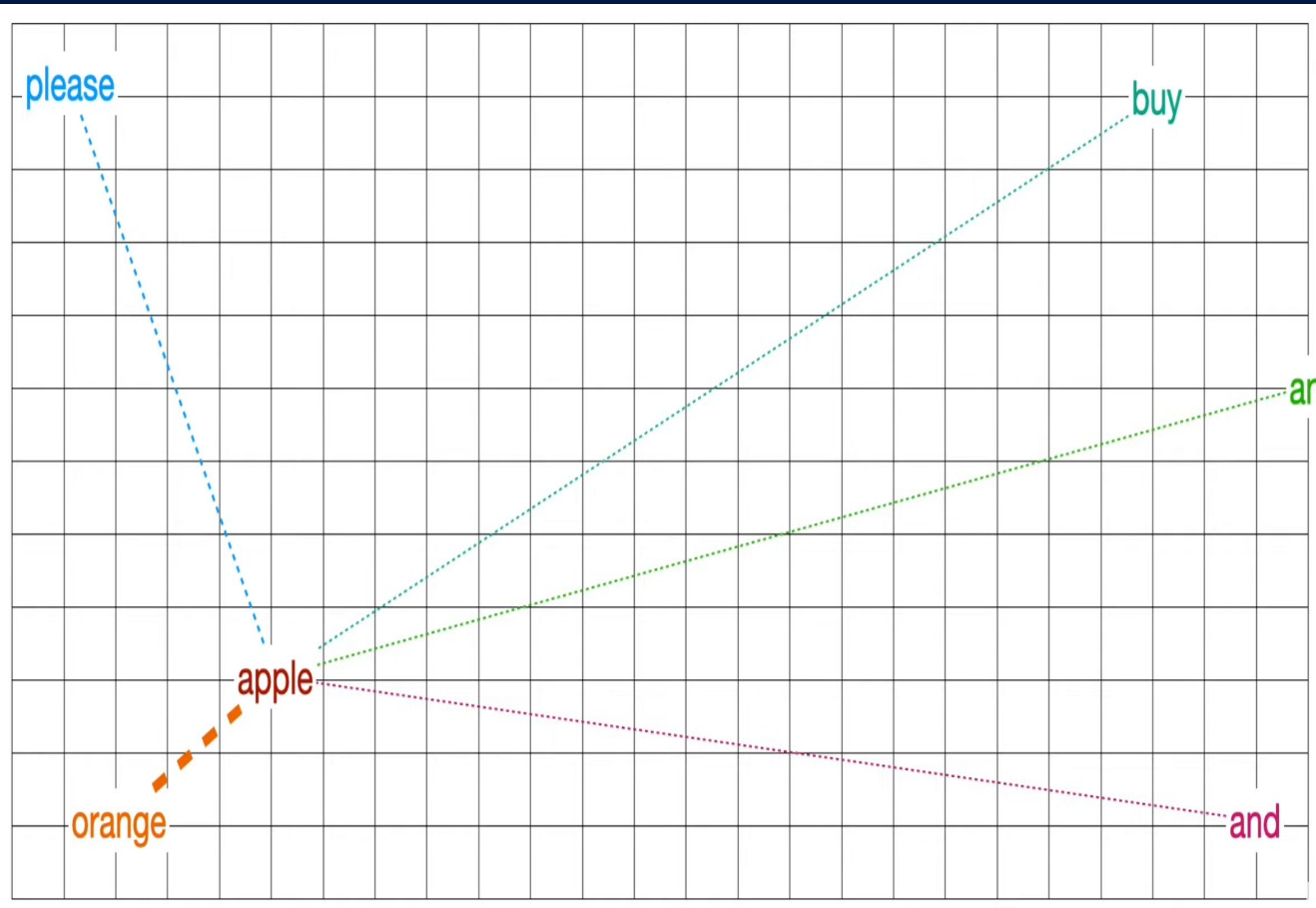


Context Engineering - Attention

Please buy an apple and an orange.

Apple unveiled a new phone.

Context Engineering - Attention - Gravity



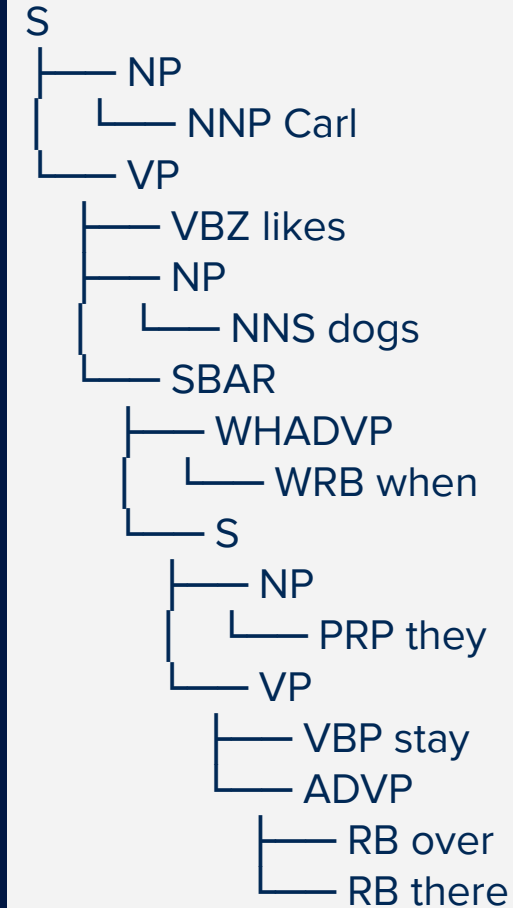
*Please buy an apple
and an **orange**.*

The more similar the vectors of two words are the more force they exert on each other.

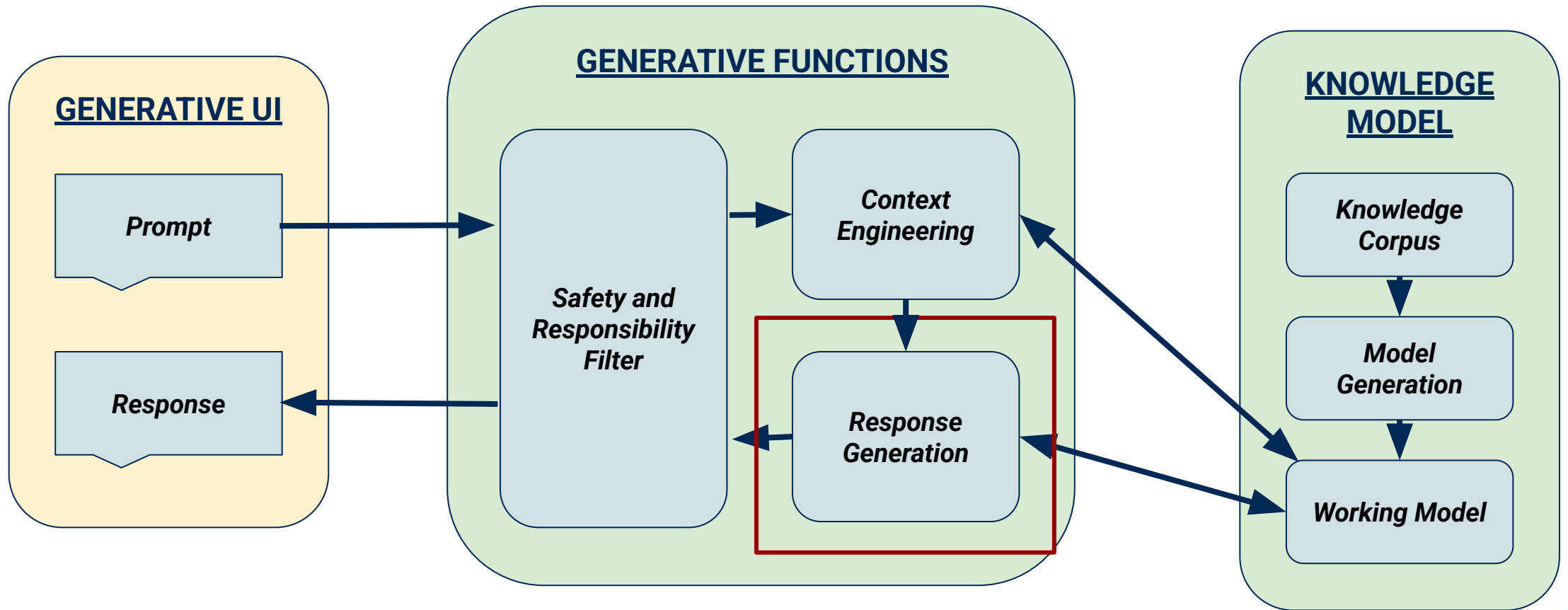
Context Engineering - Attention - NLP

Carl likes dogs when they are over there.

The known grammar and syntax of natural (and semi-natural) languages provides information that affects the “distance” between and semantic weight of words.

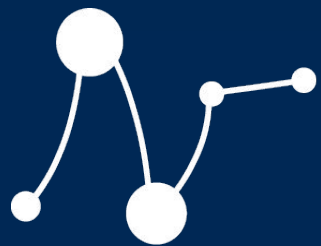


Response Generation



- Attention
- Executive Function
- [Language]
- Social Cognition

- Learning and Memory
- [Language]



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Questions?

